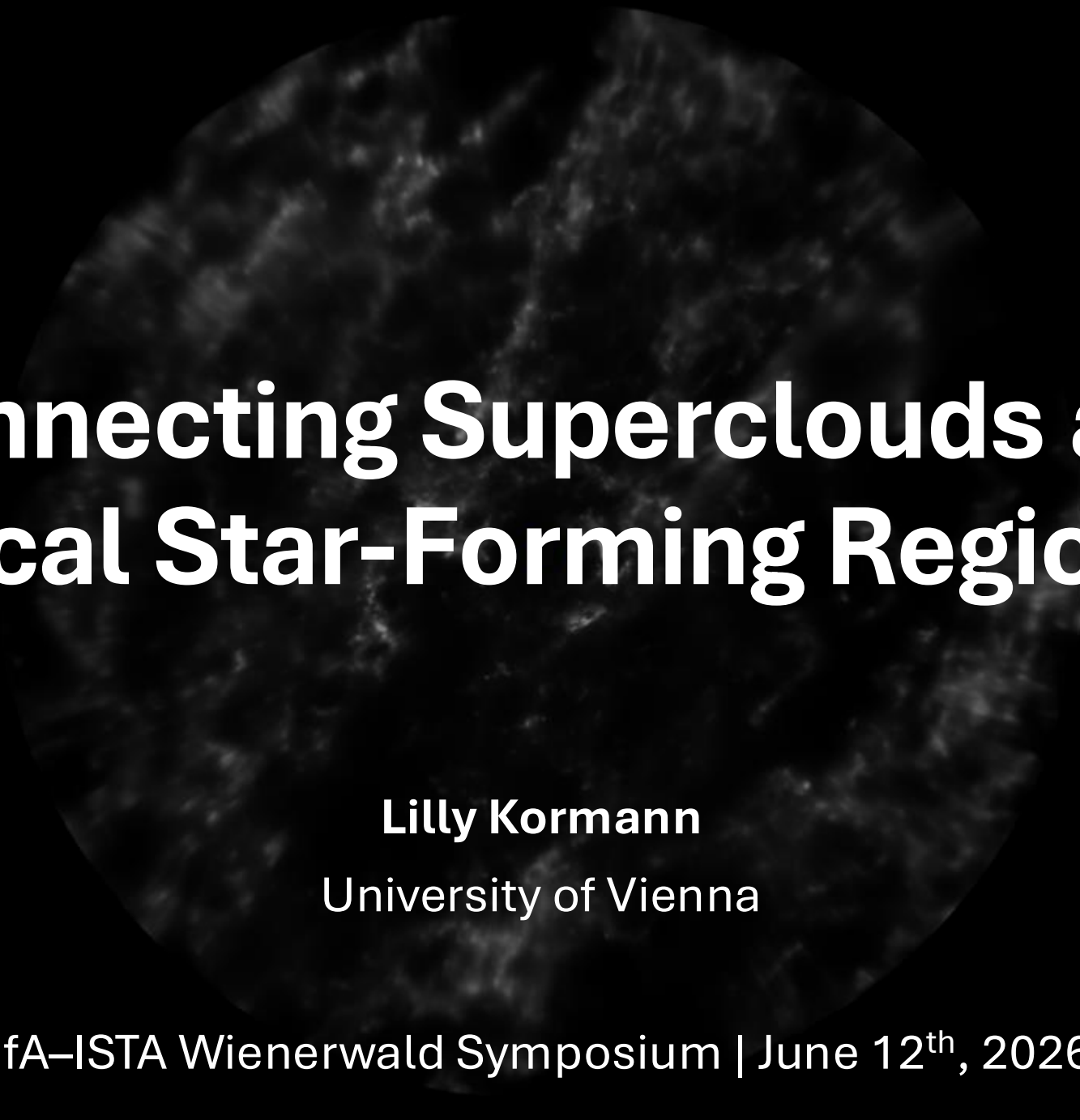




Connecting Superclouds and Local Star-Forming Regions

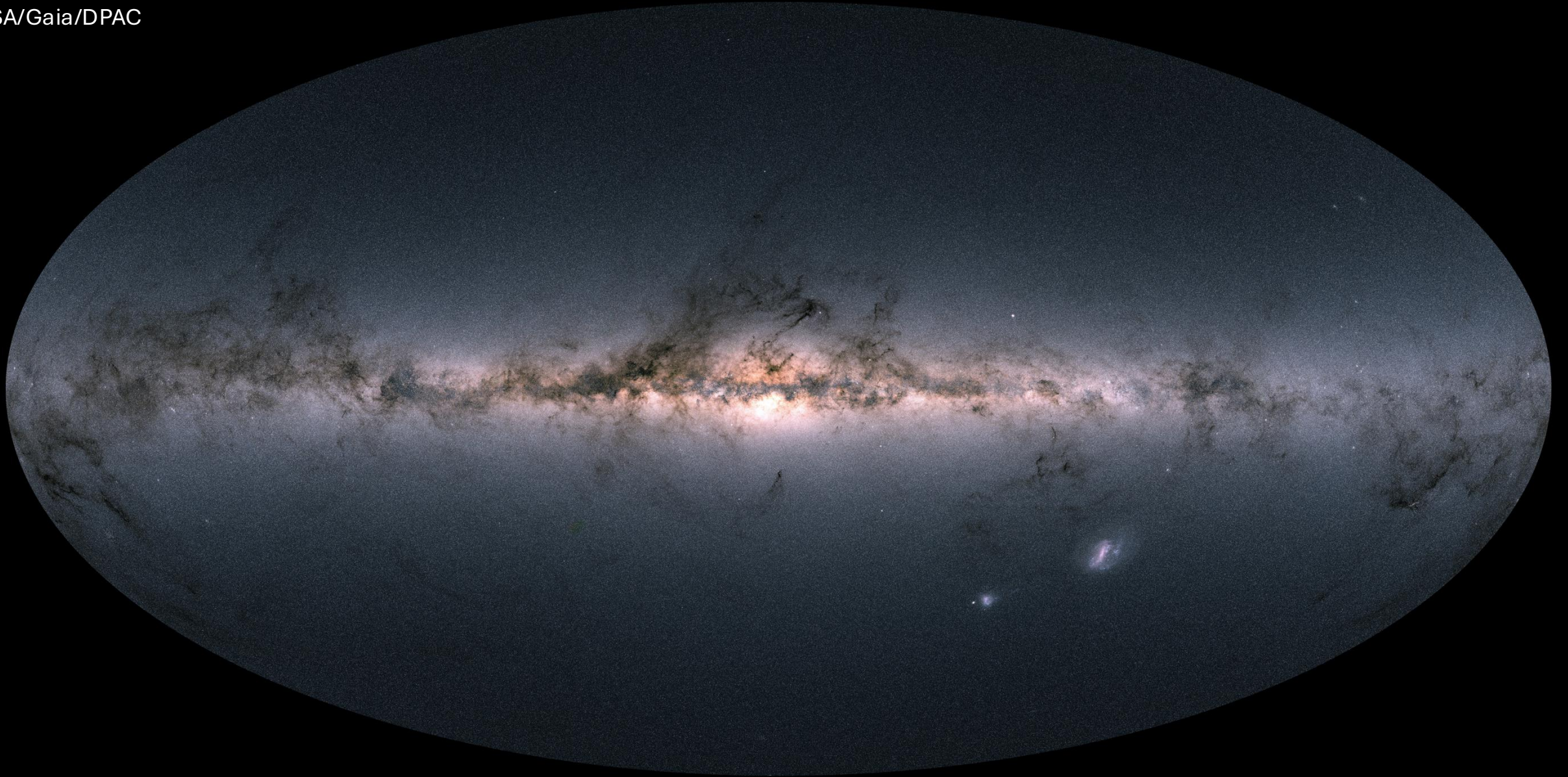
Lilly Kormann

University of Vienna

IfA–ISTA Wienerwald Symposium | June 12th, 2026

Gaia's All-Sky View

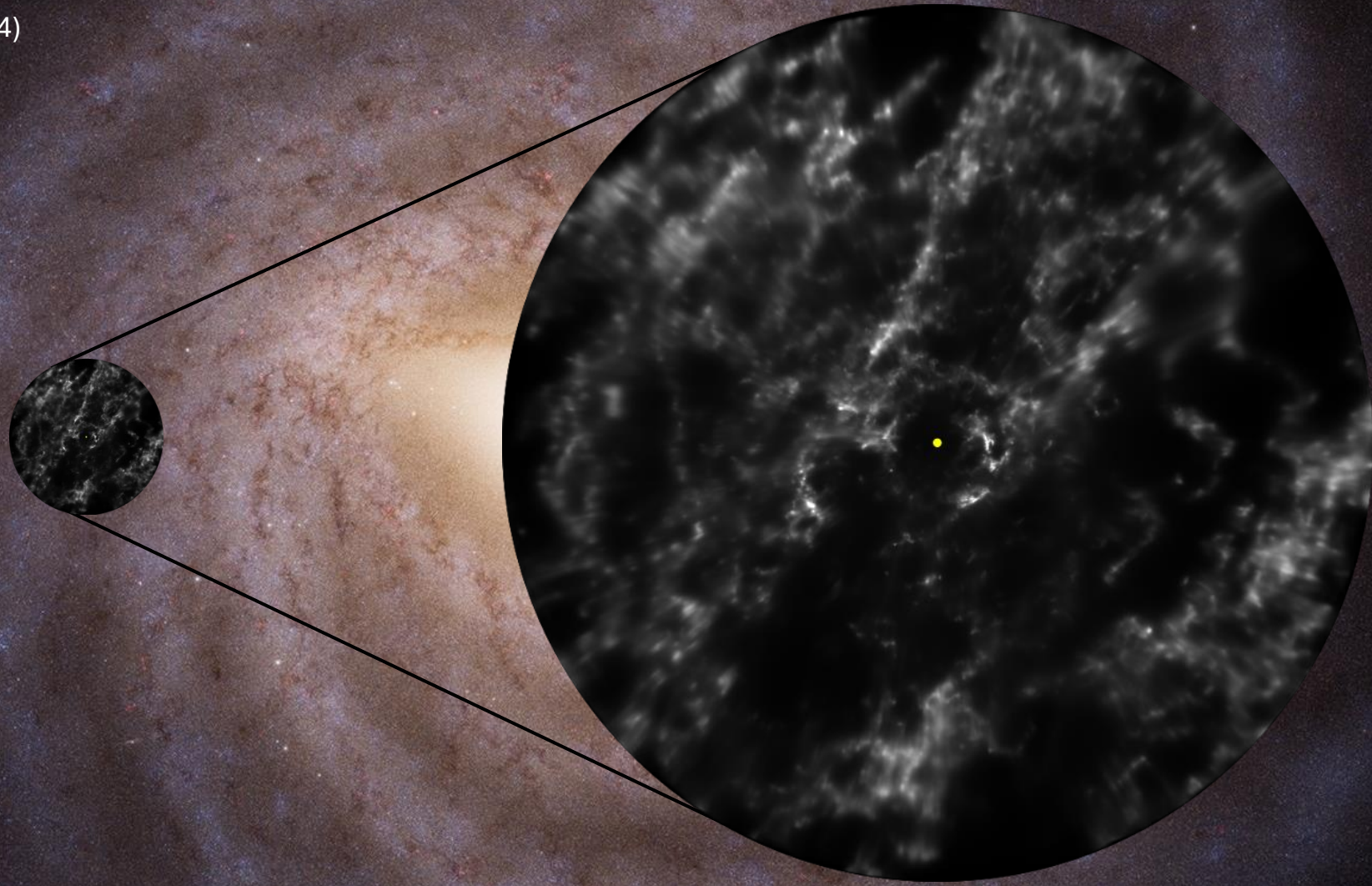
ESA/Gaia/DPAC



Artistic Impression of the Milky Way

ESA/Gaia/DPAC, Stefan Payne-Wardenaar

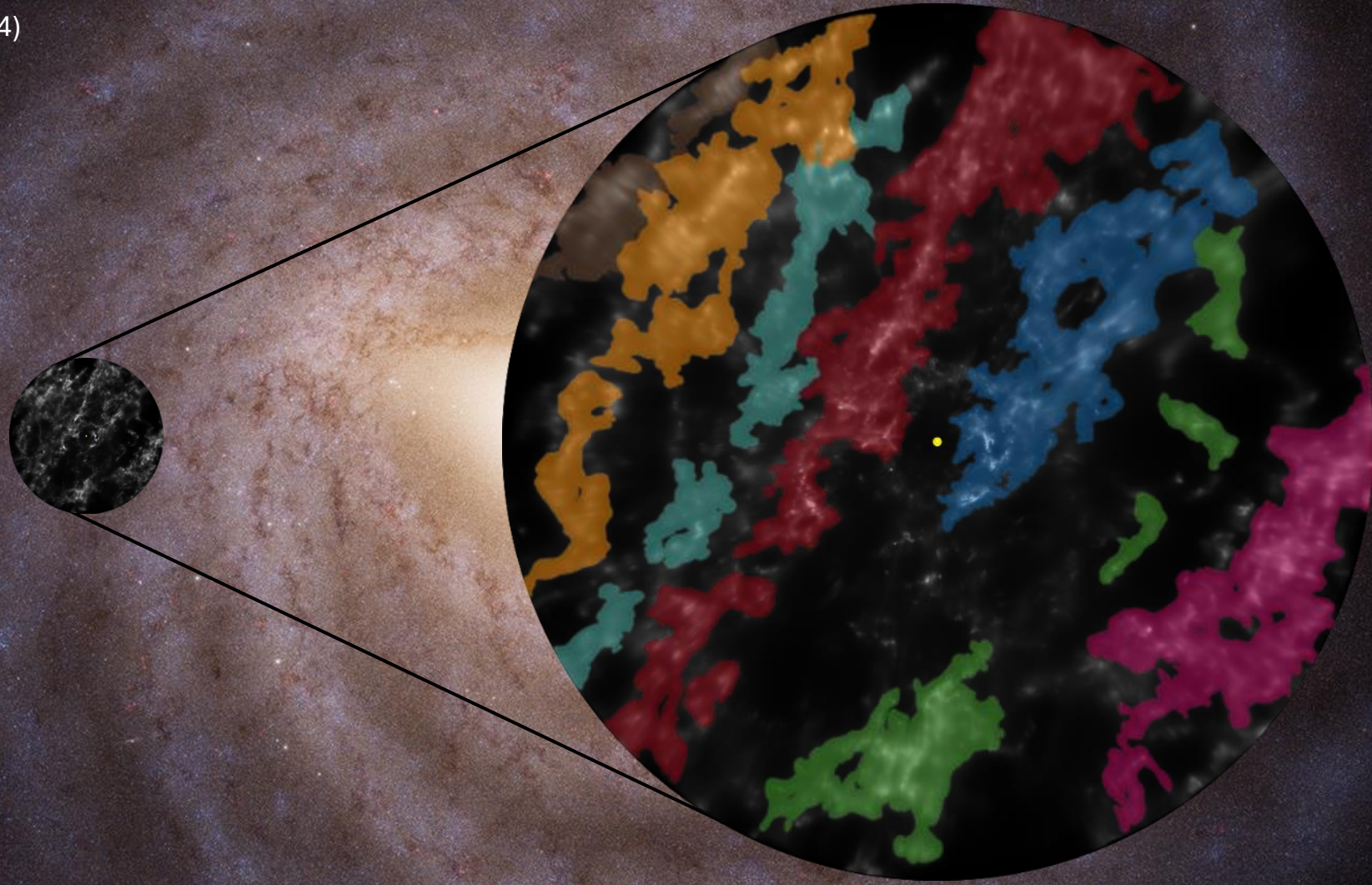
Dust map: Edenhofer et al. (2024)



Artistic Impression of the Milky Way

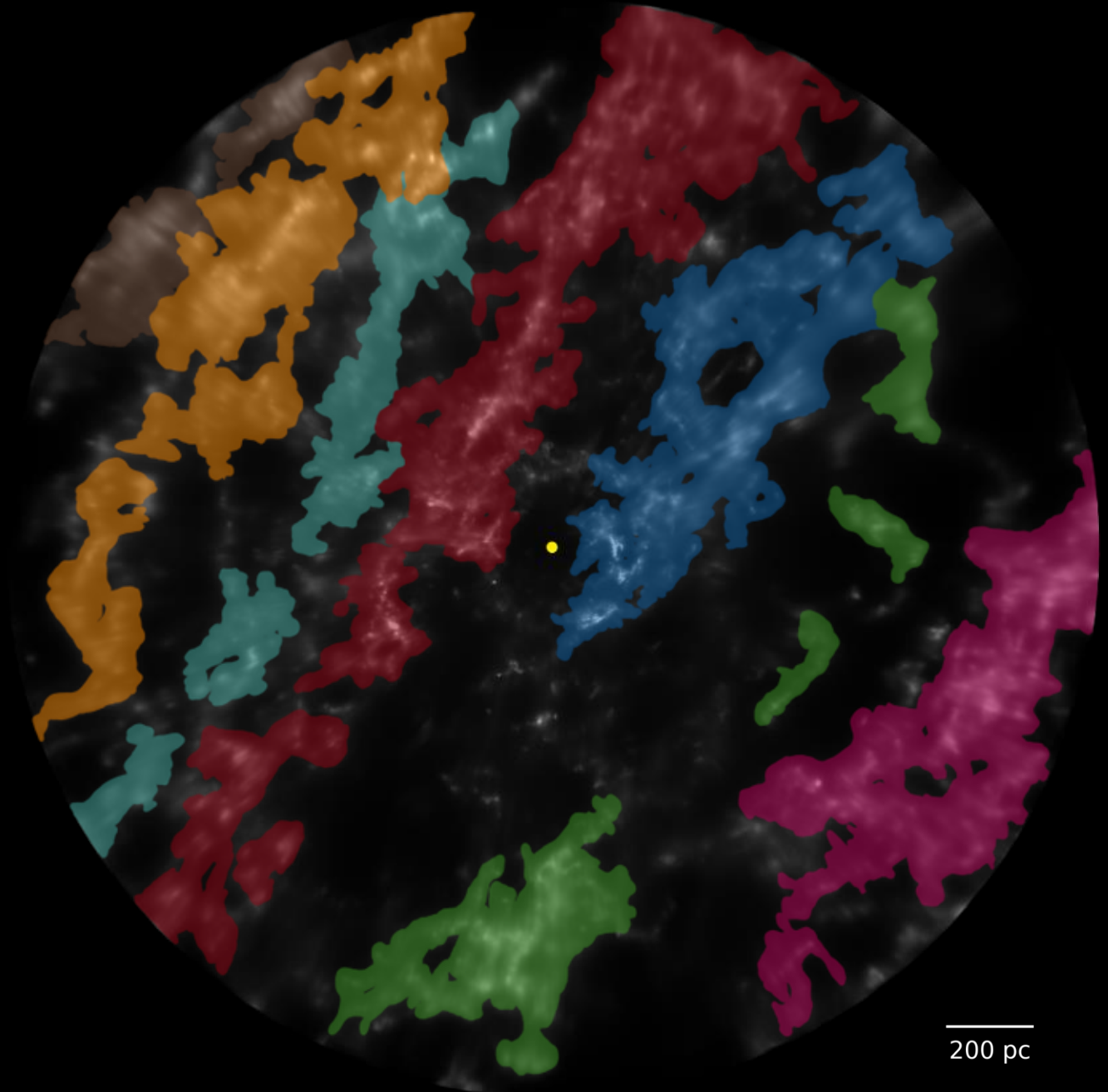
ESA/Gaia/DPAC, Stefan Payne-Wardenaar

Dust map: Edenhofer et al. (2024)



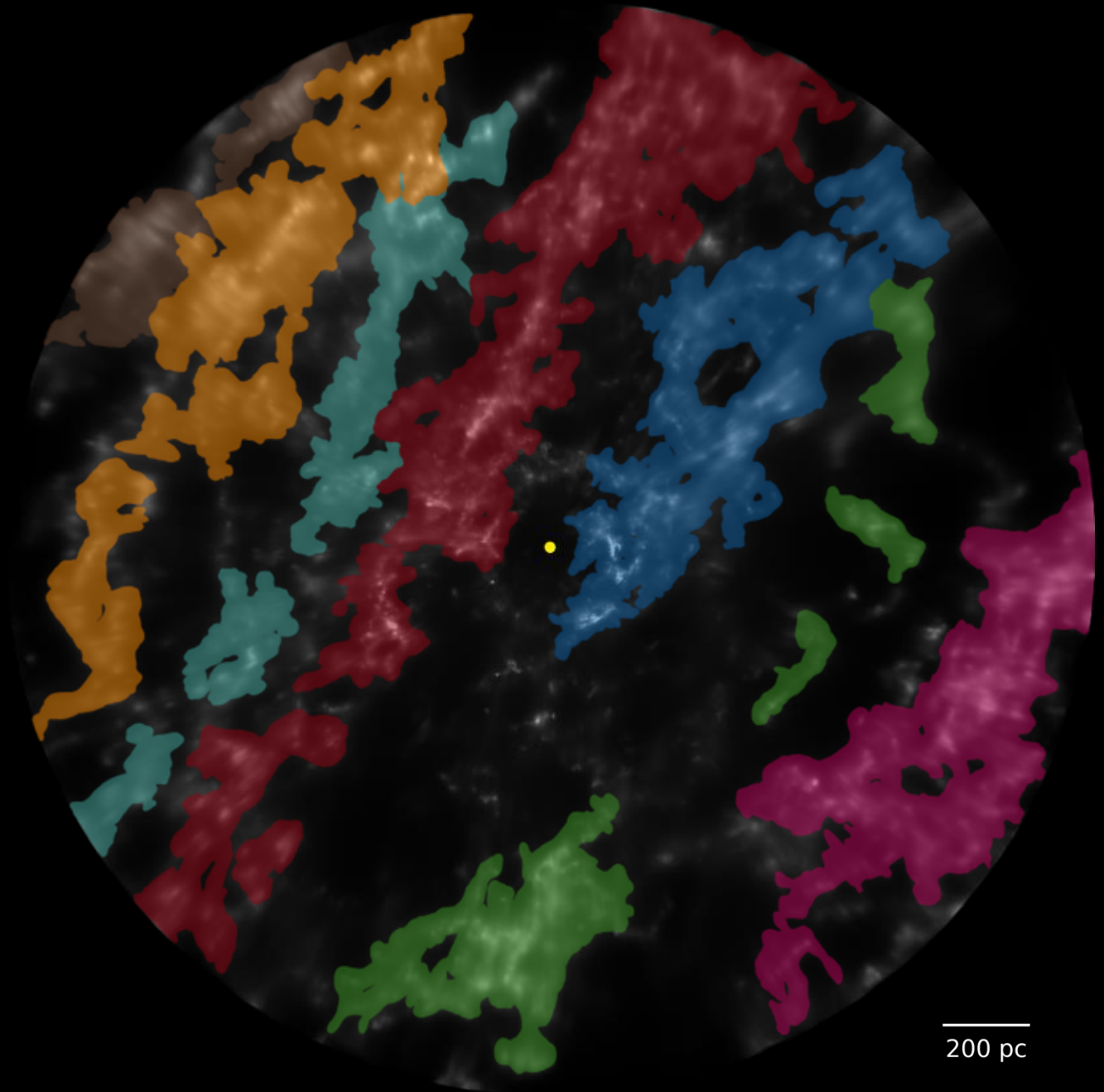
Superclouds

The seven superclouds of the local Milky Way as defined in Kormann et al. (2026).



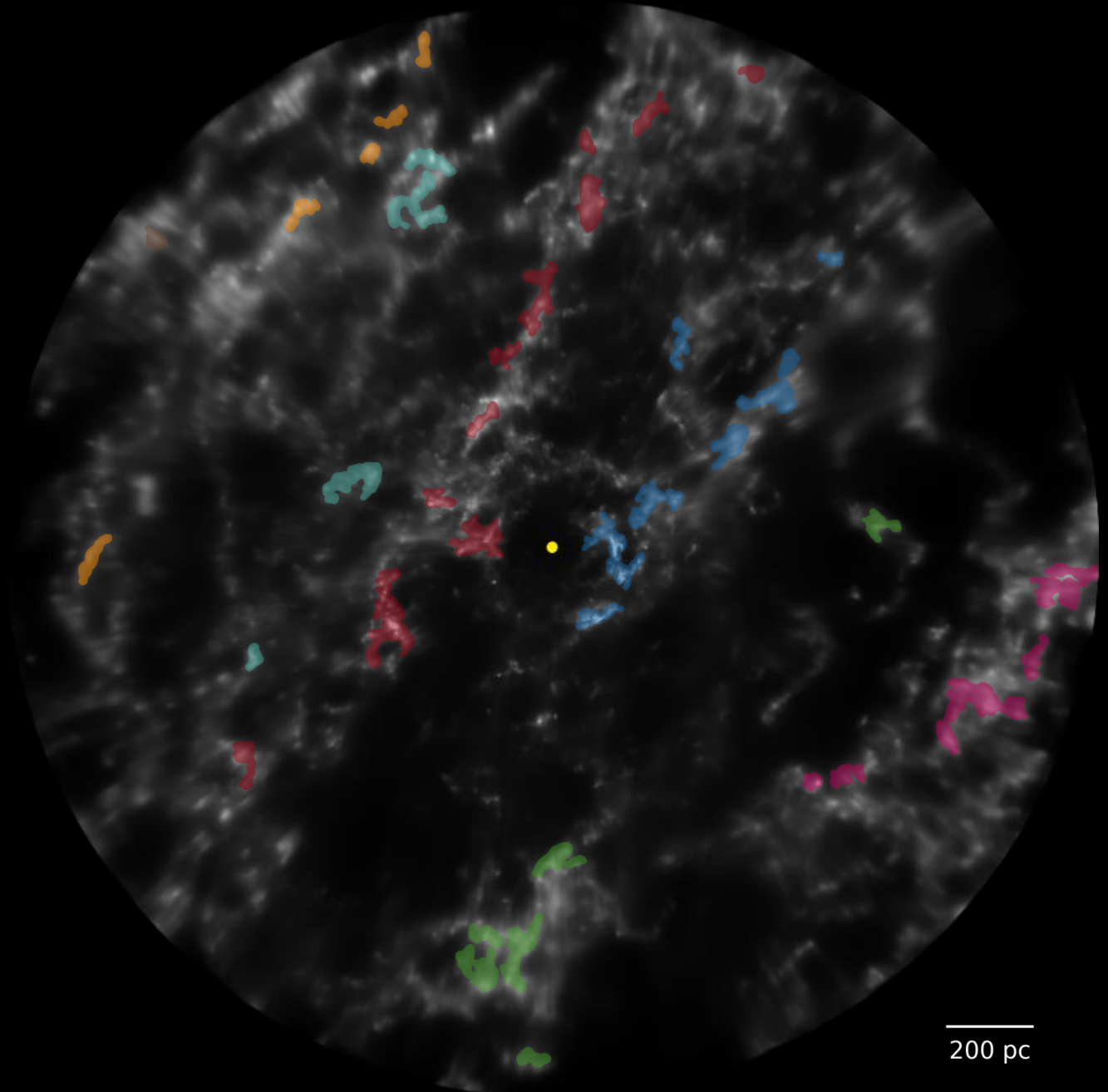
Superclouds

Segmentation done using the
HOP structure finder by
Colman et al. (2024).



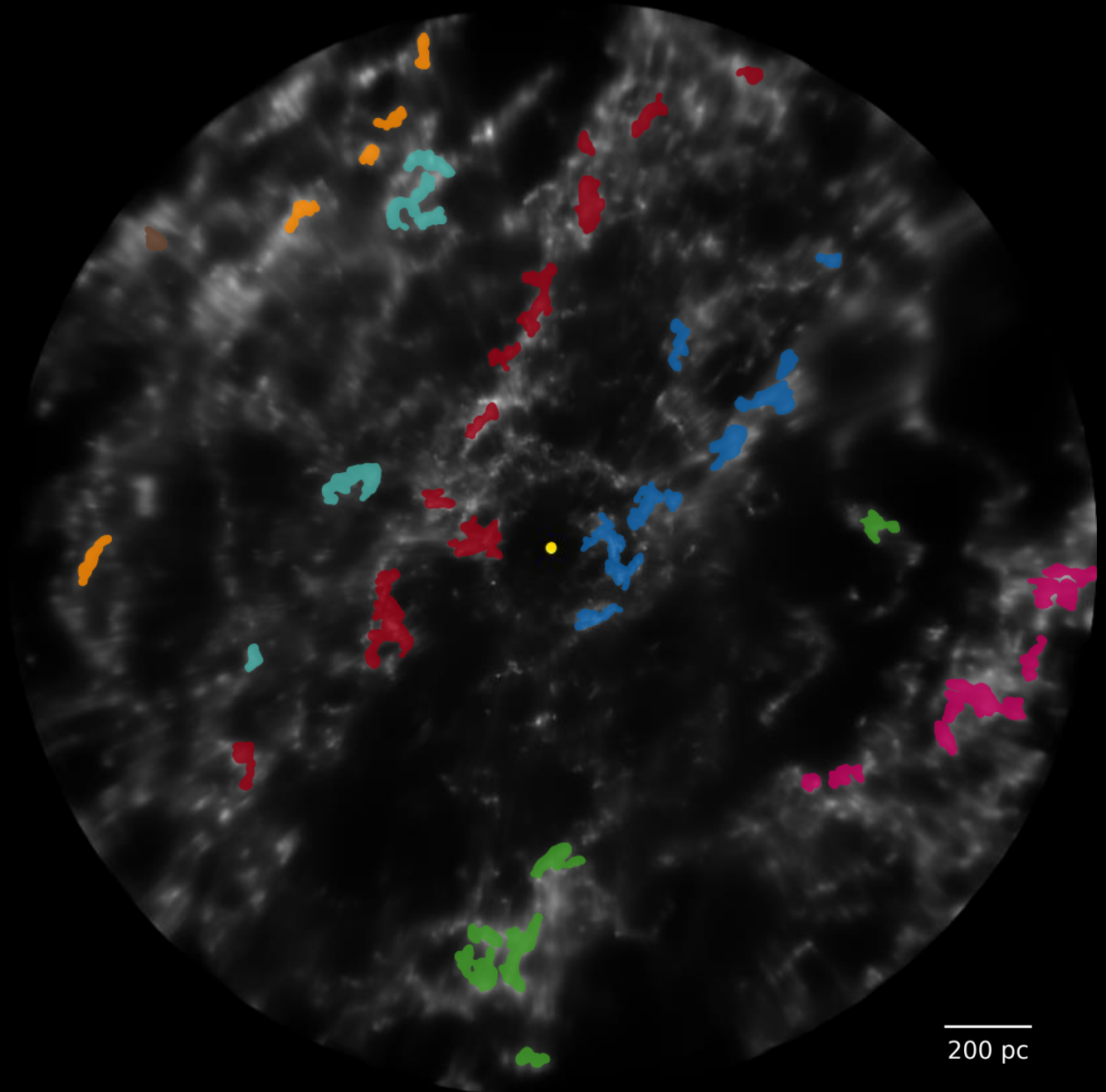
Superclumps

Regions of enhanced densities
along the spines of the
superclouds.



Superclumps

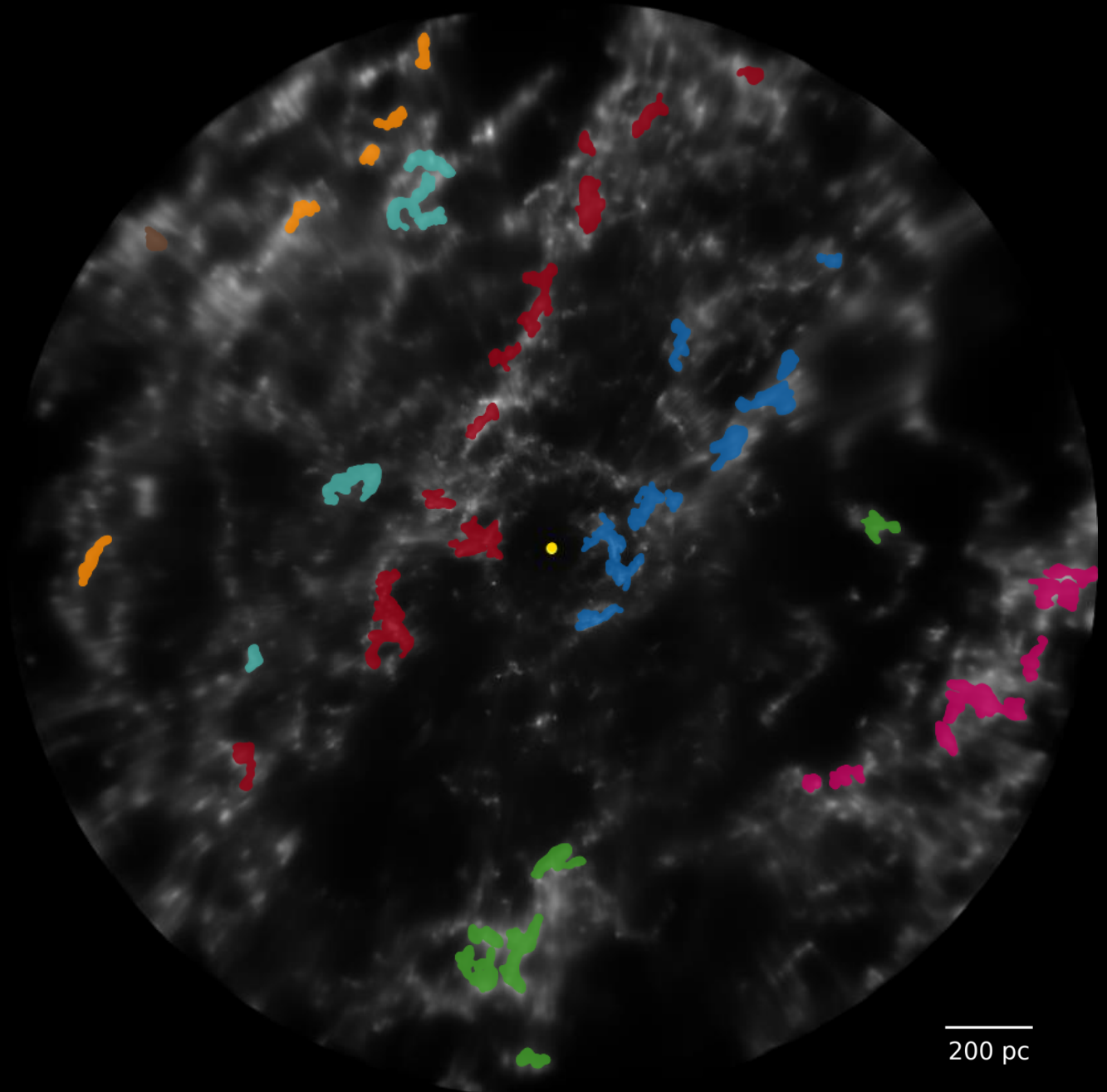
Regions of enhanced densities
along the spines of the
superclouds.



Superclumps

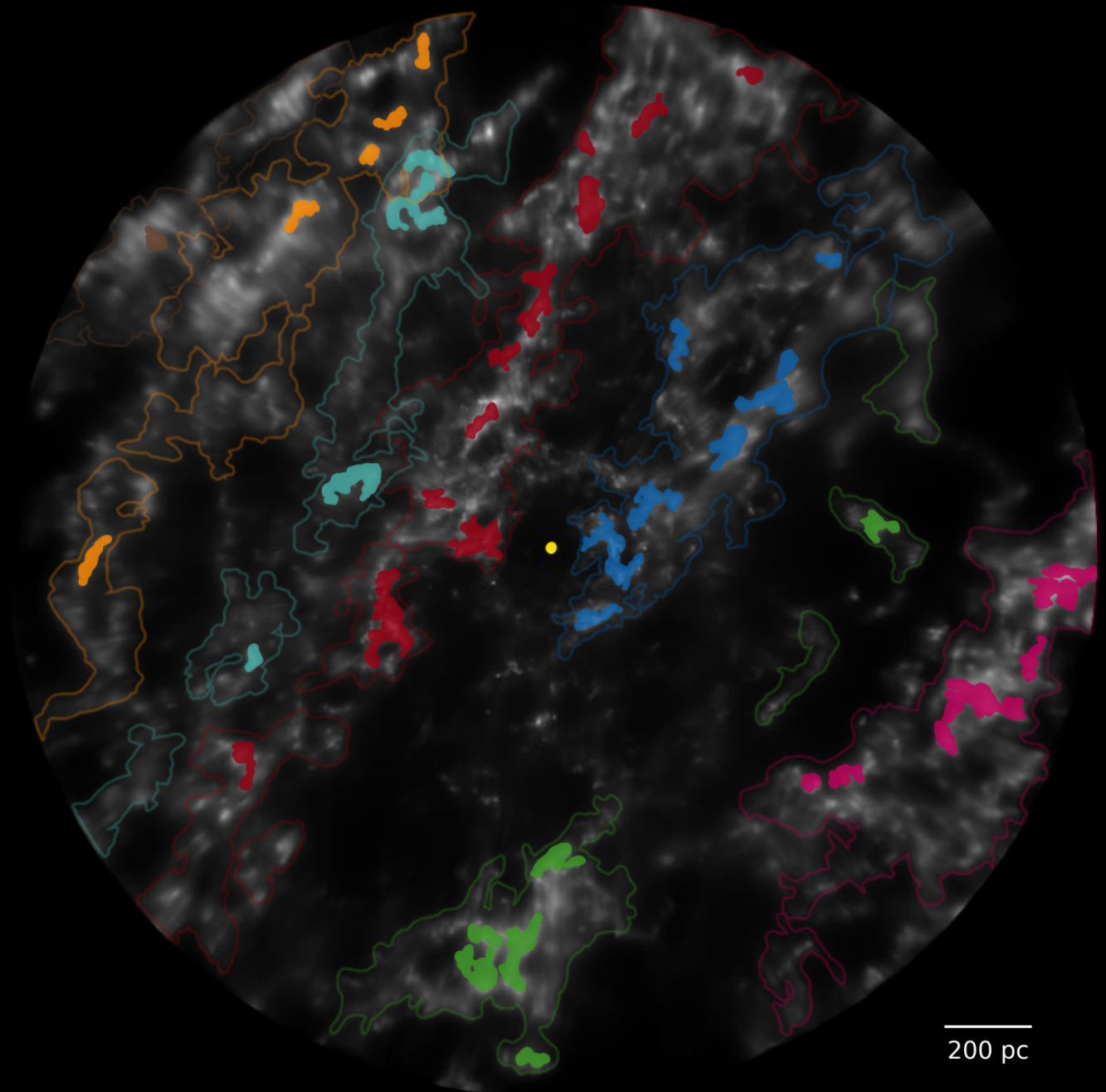
Regions of enhanced densities along the spines of the superclouds.

1. alignment with spines of the superclouds
2. correlation of superclumps and star-forming regions
3. superclump spacing



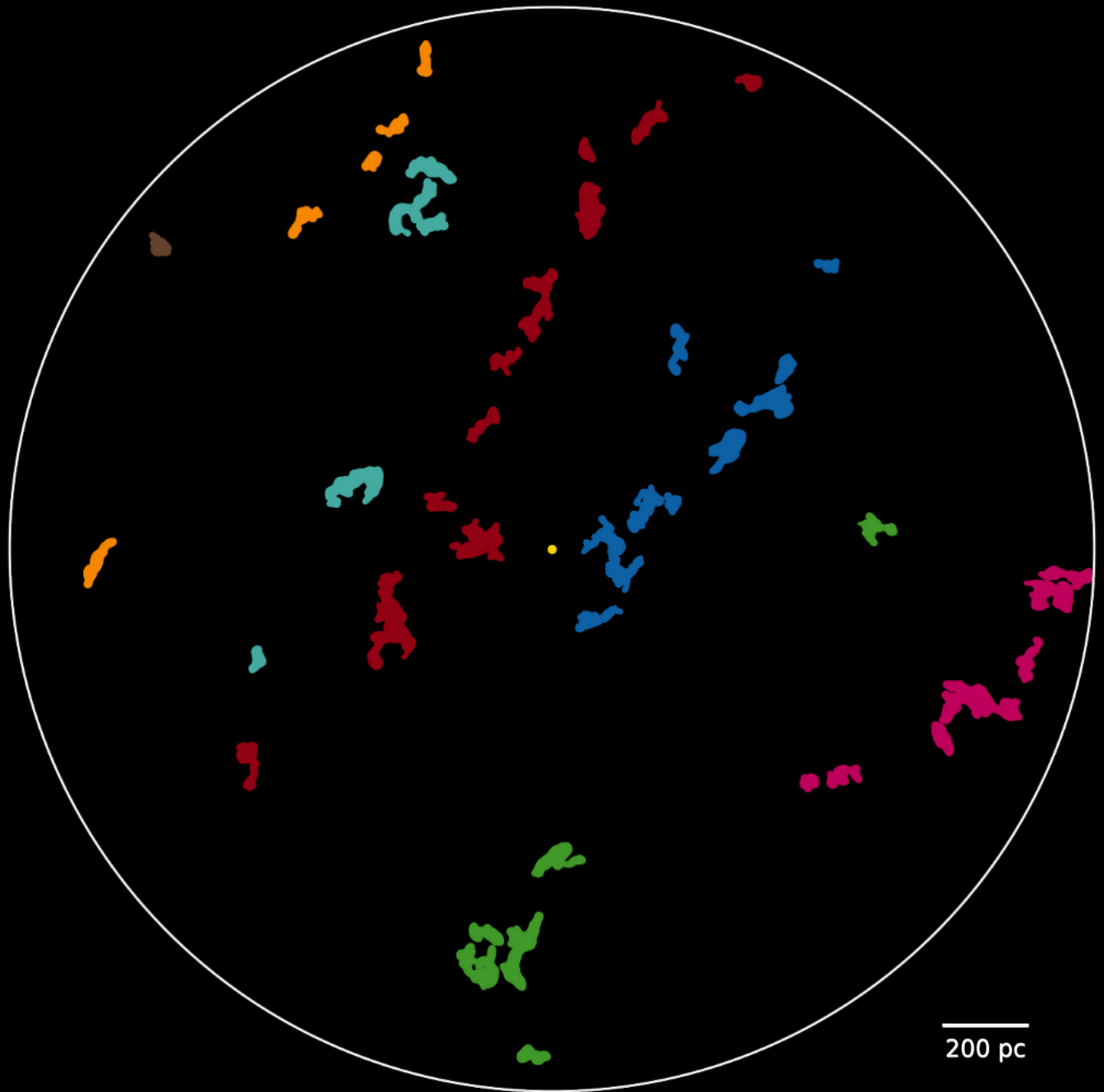
1. Alignment

We find that the superclumps are aligned with the spines of the parent supercloud.



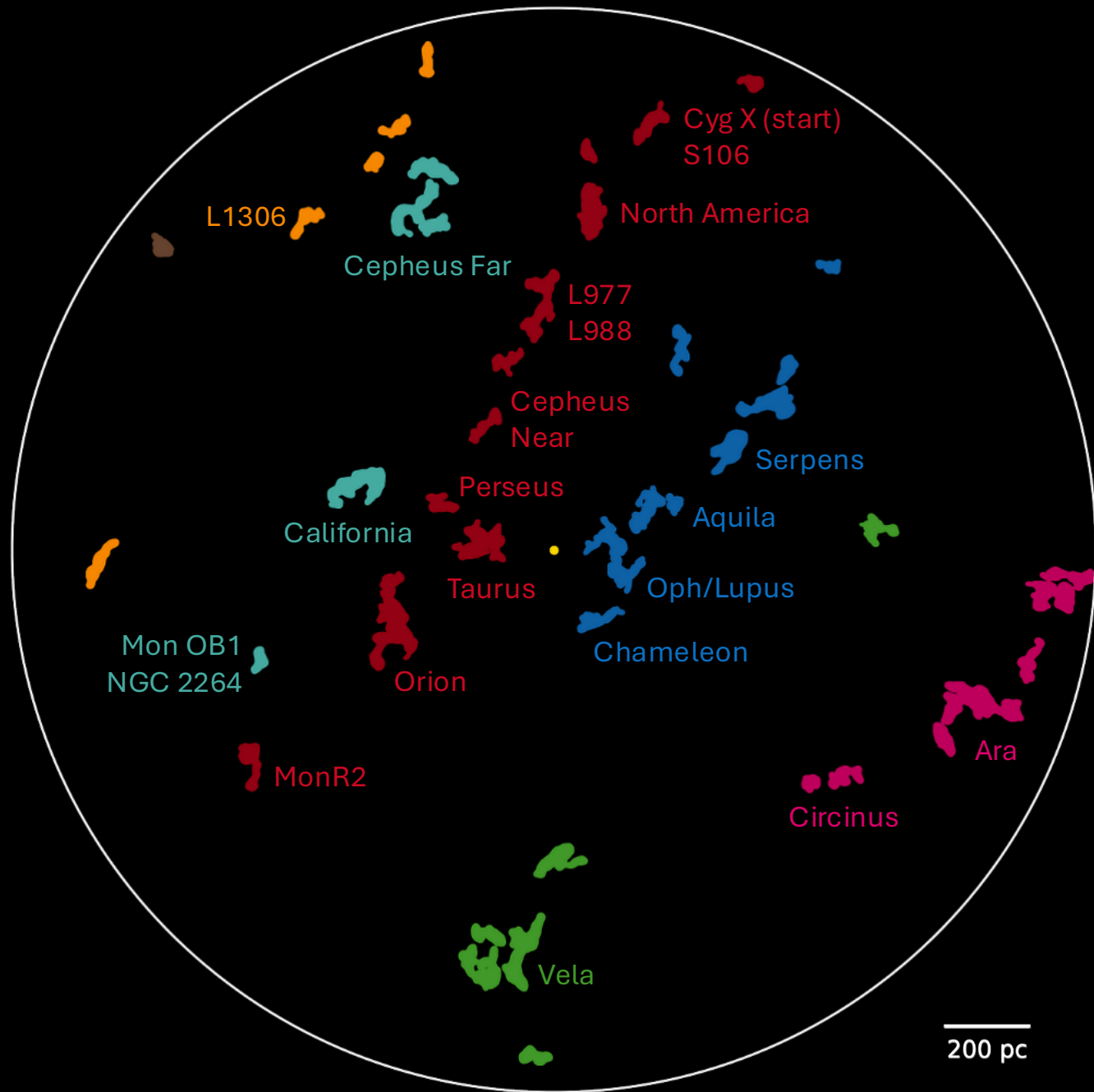
2. SFR Connection

73% of the star-forming regions in the solar neighborhood can be associated with the superclumps.



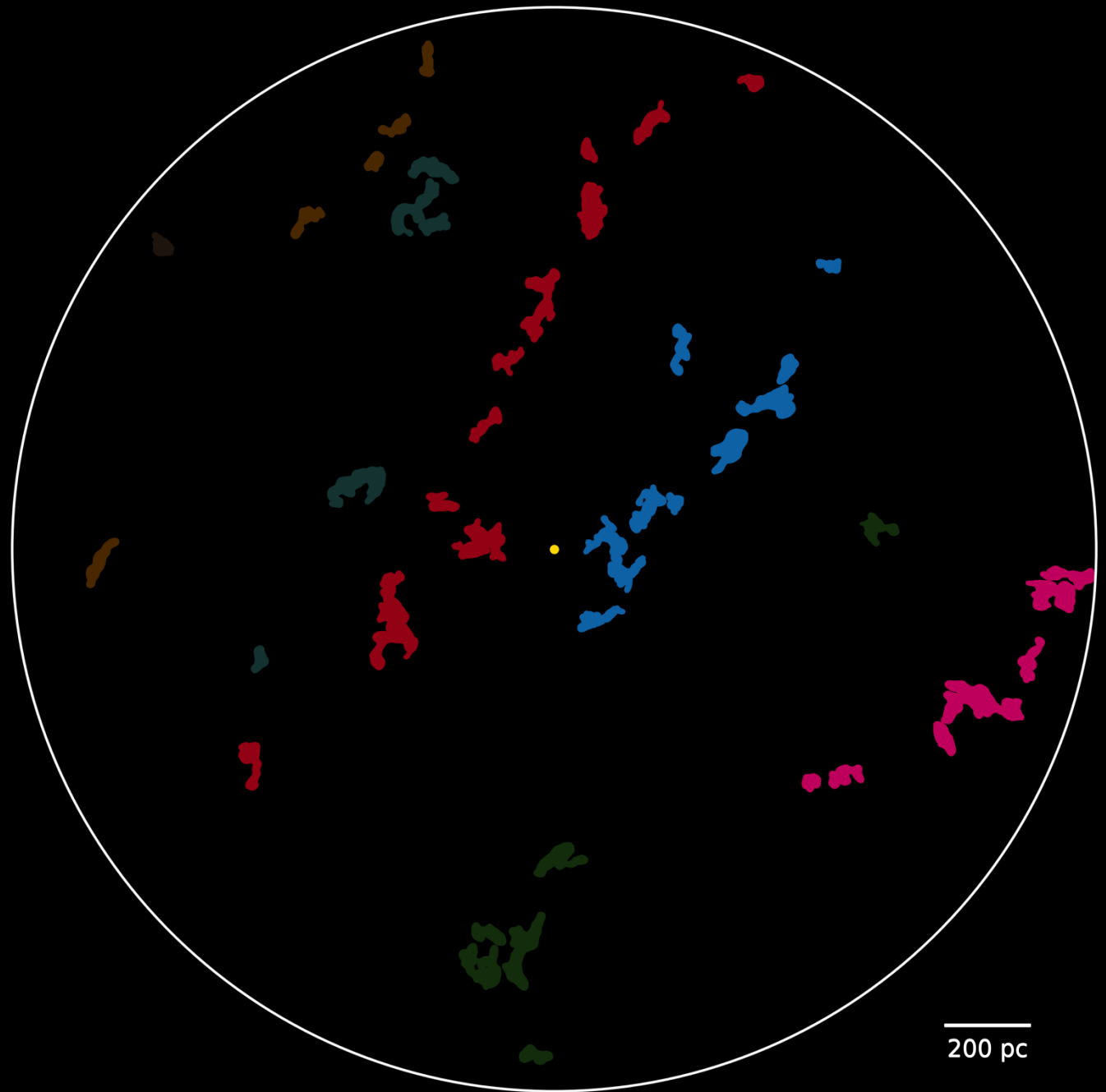
2. SFR Connection

73% of the star-forming regions in the solar neighborhood can be associated with the superclumps.



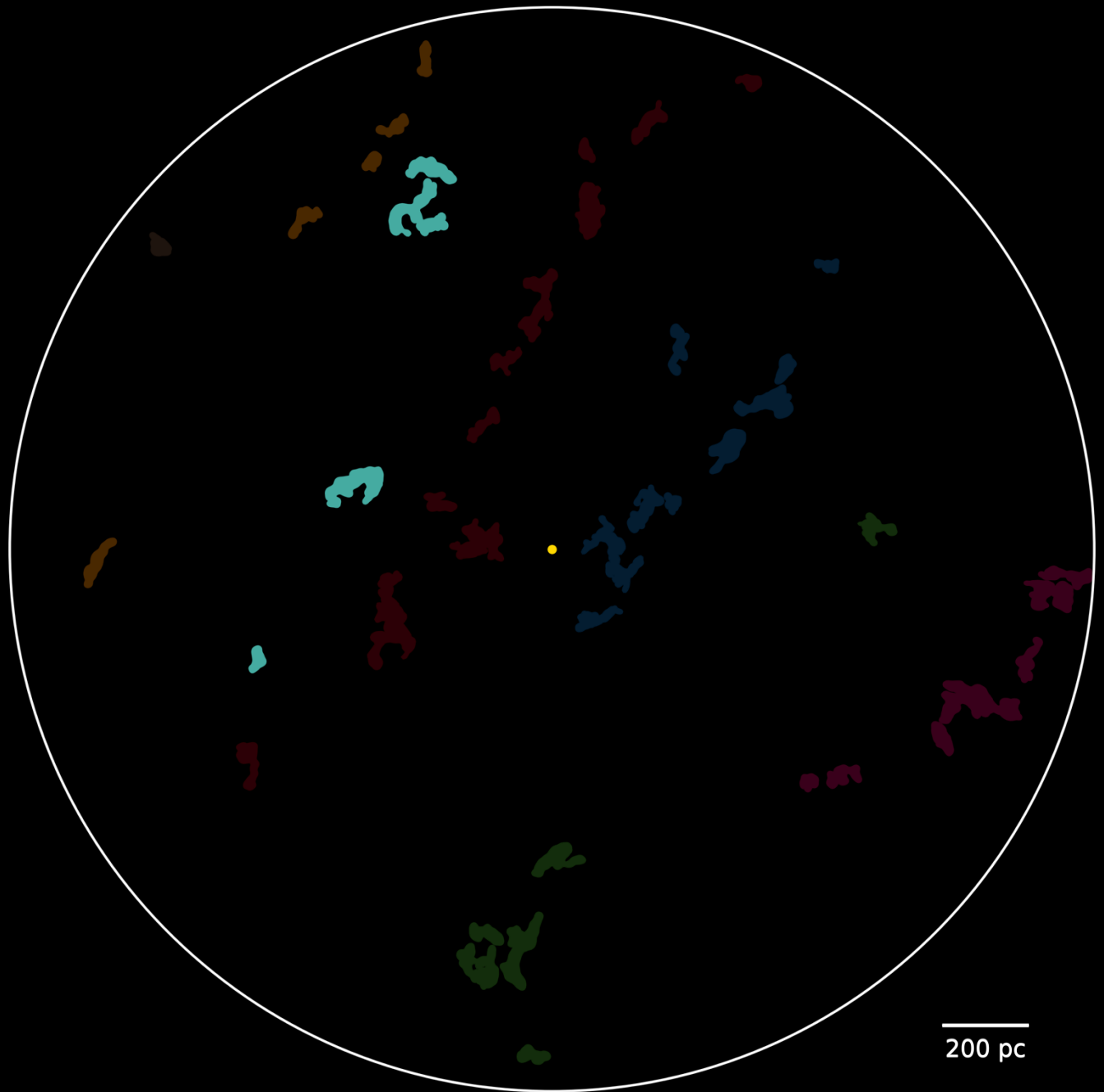
3. Spacing

The three most massive superclouds show average superclump spacings of 180 – 200 pc.



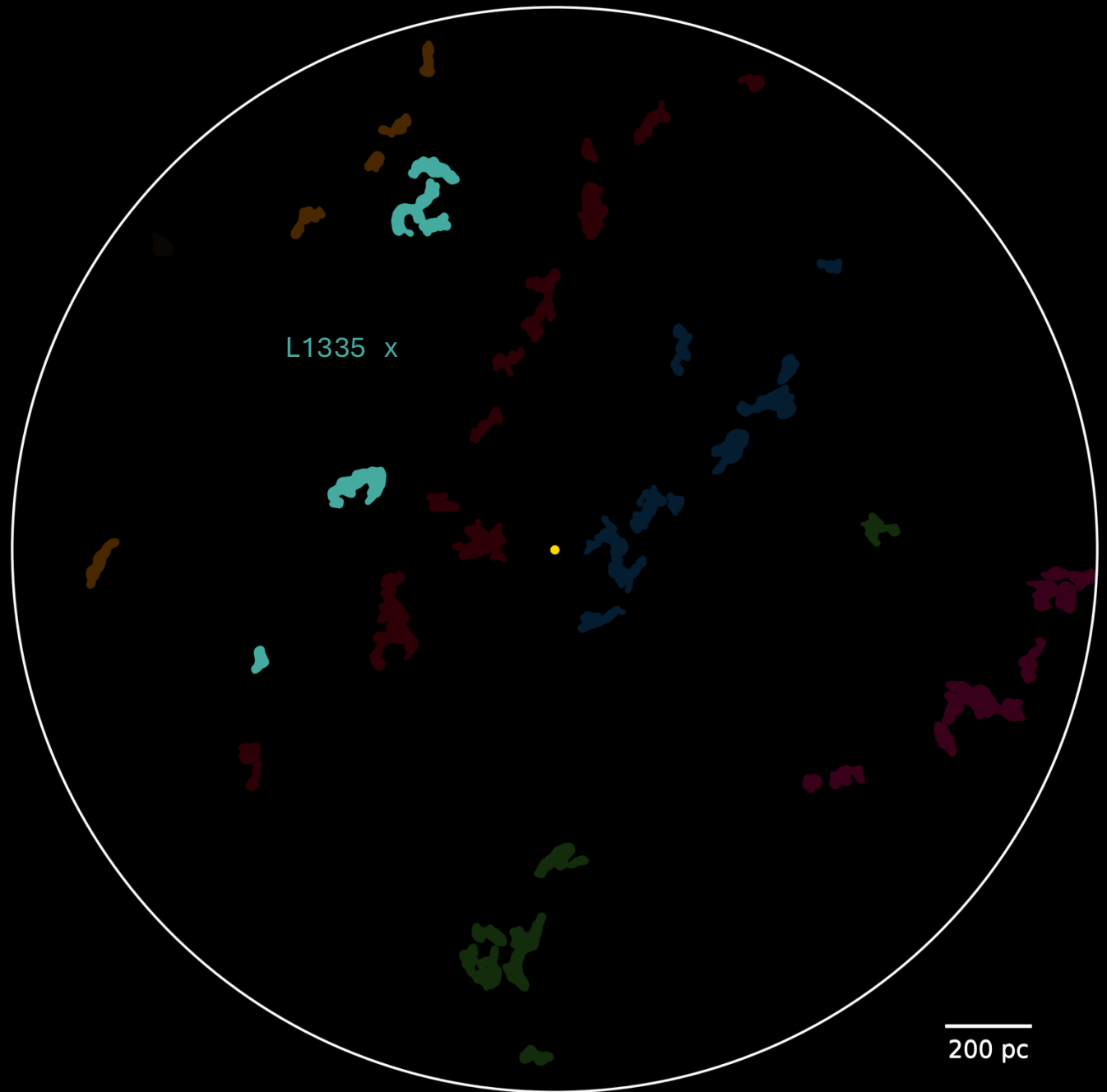
3. Spacing

The superclumps in the Natrix Cloud have an average spacing of 250 pc.



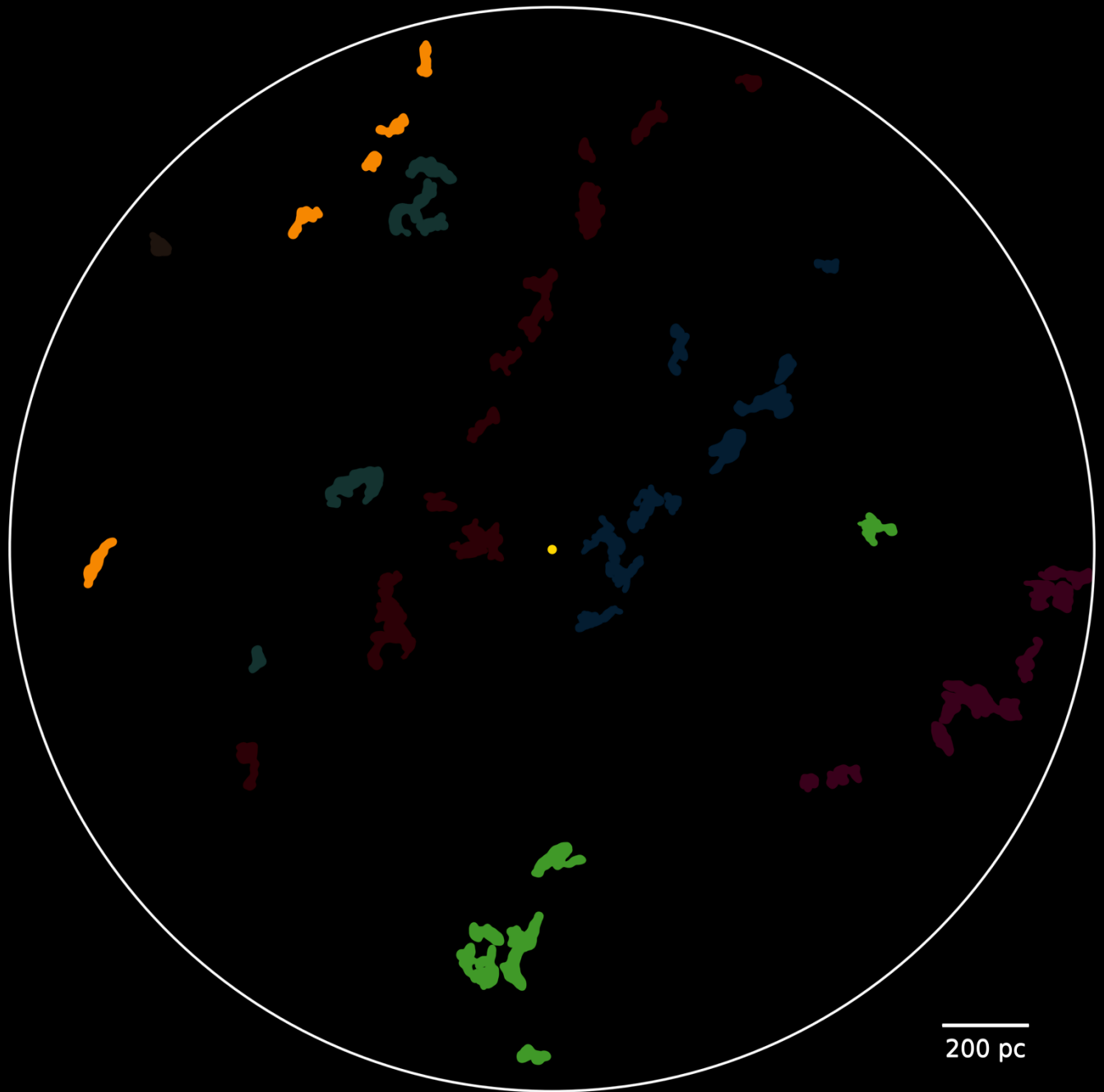
3. Spacing

Including L1335 for the distance calculation for Matrix reduces the average spacing to ~140 pc.



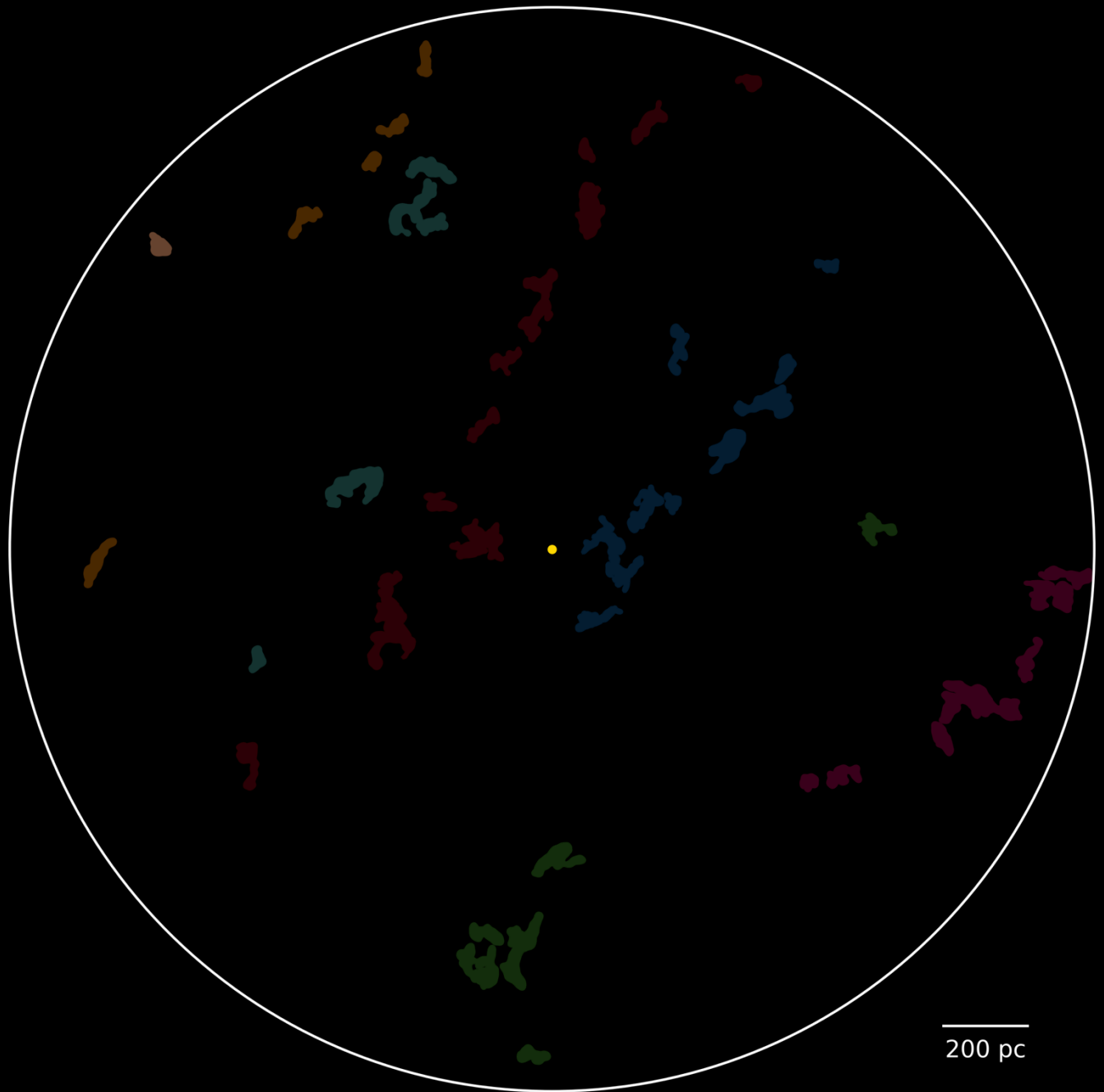
3. Spacing

Work in progress: large
superclump spacing for the
Malpolon and Vela Ridge
Cloud.



3. Spacing

No associated star-forming regions with the Anguis Cloud



Take Aways

1. We find regions of enhanced densities, termed superclumps, along the spines of the local superclouds.
2. In the solar neighborhood, 73% of known star-forming regions are associated with the superclumps.
3. The three largest superclouds have a semi-regular superclump spacing of 180 – 200 pc.

Kormann et al. 2026

